

BIL 331/BIL 531
Analysis of Algorithms

Asymptotics Review

--Read CLRS Chapters 1- 3

Asymptotic Notation

- Big O , Ω and Θ and the little ones ω and o
- Provides a way to simplify analysis
 - Allows us to ignore constant factors
 - Focus on the large values of n , focus on the dominant term

Polynomial: each term is a power of n
such as $n^5 + n^2$ or $\sqrt{n} = n^{1/2}$

Polylogarithmic: powers of $\log n$ $(\log_2 n)^5 = \log_2^5 n$

Exponential: A constant ² $a \neq 1$ a^n

How do these functions grow?

- $f_1(n) = 43n^2 \log^4 n + \boxed{12n^3 \log n} + 52n \log n \quad \Theta(n^3 \log n)$

- $f_2(n) = \boxed{15n^2} + 7n \log^3 n \quad \Theta(n^2)$

- $f_3(n) = 3n + 4 \log_5 n + \boxed{91n^2} \quad \Theta(n^2)$

- $f_4(n) = 13 \cdot \underline{3^{2n+9}} + 4n^9 \quad \Theta(9^n)$

- $f_5(n) = \sum_{i=0}^{\infty} \frac{1}{2^i} = 1 + \frac{1}{2} + \frac{1}{2^2} + \dots = \frac{1}{1 - \frac{1}{2}} = 2 \in \Theta(1)$

f_1 is a slower algorithm than f_2

same
growth
rate

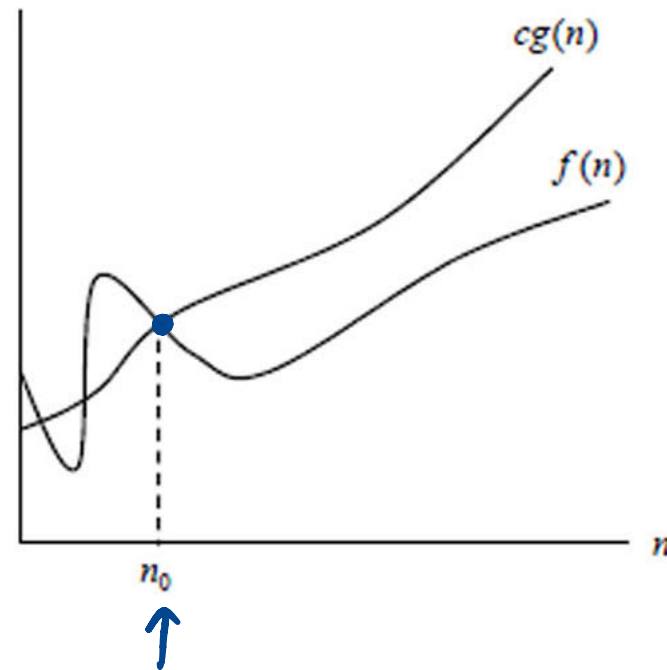
Big O

$$f(n) \in O(g(n))$$

- $f(n) = O(g(n))$ if there exists constants $n_0 \geq 0, c > 0$ such that

$$f(n) \leq c \cdot g(n), \quad \forall n \geq n_0$$

- Upper bound



Big Θ

$f(n) \overset{c}{=} \Theta(g(n))$ if there exists constants $n_0 \geq 0, c_1 > 0$,
and $c_2 > 0$ such that

$$\underbrace{c_1 \cdot g(n)}_{\text{lower bound}} \leq f(n) \leq \underbrace{c_2 \cdot g(n)}_{\text{upper bound}}, \quad \forall n \geq n_0,$$

- Tight bound

if $f(n) \in \Theta(g(n))$ then
 $\hookrightarrow f(n) \in O(g(n))$
 $\hookrightarrow f(n) \in \Omega(g(n))$

Big Ω

$f(n) = \Omega(g(n))$ if there exists constants $n_0 \geq 0, c > 0$ such that

• Lower bound $\underbrace{f(n) \geq c \cdot g(n)}_{\text{lower bound}}, \quad \forall n \geq n_0$

$$f(n) = 5n^2 + 3n - 4 \stackrel{?}{\in} \underline{\Theta}(n^2)$$

$\uparrow$
 $g(n)$

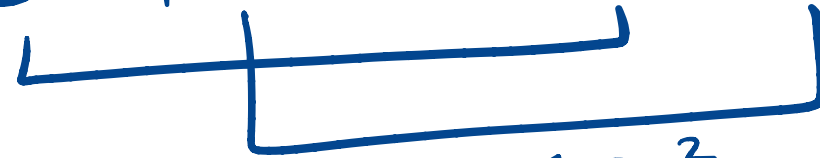
Example

Upperbound

$$5n^2 + 3n - 4 \leq c_2 n^2 \quad n \geq n_0$$

$$5n^2 + 3n - 4 \leq 5n^2 + 3n^2 = \boxed{8} n^2$$

c_2



$$3n \leq 3n^2$$

$$\boxed{1} \leq n$$

n_0

$$c_2 = 8$$

$$n_0 = 1$$

Lowerbound

pick $c_1 = 5$

$$c_1 n^2 \leq 5n^2 + 3n - 4$$

$$n \geq n_0$$

$$\cancel{5n^2} \leq \cancel{5n^2} + 3n - 4$$

$$0 \leq 3n - 4$$

$$\boxed{\frac{4}{3}} \leq n$$

n_0

pick max n_0
 $n_0 = \frac{4}{3}, c_1 = 5,$
 $c_2 = 8$

what if $f(n) = 5n^2 - 3n + 2$

Lowerbound
pick $c_1 = 5$

$$\cancel{5n^2} \leq \cancel{5n^2} - 3n + 2$$

$$3n \leq 2$$

$$\boxed{n \leq \frac{2}{3}}$$

$c_2 = 5$ does not work

try $c_1 = 4$

$$4n^2 \leq 5n^2 - 3n + 2$$

$$0 \leq n^2 - 3n + 2$$

$$0 \leq \underbrace{(n-2)(n-1)}_{\geq 0}$$

if $n \geq \boxed{2}_{n_0}$ $\left(\begin{array}{l} n_0 = 2 \\ c_1 = 4 \end{array} \right)$

$$f(n) \in \underline{o}(g(n)) \text{ if } f(n) < c g(n), \quad \forall n \geq n_0$$

↑
strictly smaller

$$f(n) \in \underline{O}(g(n)) \text{ if } \exists c, g(n) < f(n) \quad \forall n \geq n_0$$

Limit Rule

c is a positive constant but not infinity

Notation	Relational Form	Limit Definition
$f(n) = o(g(n))$	$f(n) < g(n)$	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$
$f(n) = O(g(n))$	$f(n) \leq g(n)$	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = c \text{ or } 0$
$f(n) = \Theta(g(n))$	$f(n) \approx g(n)$	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = c$
$f(n) = \Omega(g(n))$	$f(n) \geq g(n)$	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = c \text{ or } \infty$
$f(n) = \omega(g(n))$	$f(n) > g(n)$	$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$

Useful Facts

- Polynomial functions grow faster than polylogarithmic functions

- $\lim_{n \rightarrow \infty} \frac{(\log n)^a}{n^b} = 0, \quad a, b > 0$

- Exponentials grow faster than polynomials

- $\lim_{n \rightarrow \infty} \frac{n^a}{b^n} = 0, \quad a > 0, b > 1$

- log bases do not matter

- $\lim_{n \rightarrow \infty} \frac{\log_a n}{\log_b n} = c \neq 0, \quad a, b > 1$

- exponential bases do matter

- $\lim_{n \rightarrow \infty} \frac{a^n}{b^n} = 0, \quad 1 < a < b$

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$$\lim_{n \rightarrow \infty} \frac{2^n}{3^n} = 0$$

Example

- $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)}$
- $= \lim_{n \rightarrow \infty} \left(\frac{15n^2 + 7n \log^3 n}{n^2} \right)$
- $= \lim_{n \rightarrow \infty} \left(\frac{15n^2}{n^2} + \frac{7n \log^3 n}{n^2} \right)$
- $= \lim_{n \rightarrow \infty} \left(15 + \frac{7n \log^3 n}{n^2} \right)$
- $= 15$

$$\begin{aligned} f(n) &\in \Theta(g(n)) \\ f(n) &\in O(g(n)) \\ f(n) &\notin \Omega(g(n)) \end{aligned}$$

$f(n) <, \approx, > g(n)?$

- $f(n) = 3^{\frac{n}{2}}, g(n) = 2^{\frac{n}{3}}$

$\emptyset \rightarrow f(n) \in o(g(n))?$
 $\subset \rightarrow f(n) \in \Theta(g(n))?$
 $\infty \rightarrow f(n) \in \omega(g(n))?$

- $f(n) = \lg(n^2), g(n) = (\lg n)^2$

- $f(n) = n^{\lg 4}, g(n) = 2^{2 \lg n}$

$$f(n) = 3^{1/2}$$

$$g(n) = 2^{1/3}$$

$$f(n) = (3^{1/2})^n$$

$$= (2^{1/3})^n$$

$$= (\sqrt{3})^n$$

$$= (\sqrt[3]{2})^n$$

$$\approx 1.73^n$$

$$\approx 1.26^n$$

$$f(n) > g(n)$$

$$f(n) \in \omega(g(n))$$

$$f(n) = \lg(n^2) \quad g(n) = (\lg n)^2$$

$$= 2 \lg n$$

$$\lim_{n \rightarrow \infty} \frac{2 \lg n}{(\lg n)^2} = 0 \rightarrow f(n) < g(n)$$

$$f(n) \in o(g(n))$$

$$f(n) = n^{\lg 4}$$

$$= n^2$$

$$g(n) = 2^{2 \lg n}$$

$$= 2^{\lg n^2}$$

$$= (n^2)^{\lg 2} = n^2$$

$$f(n) \in \Theta(g(n))$$

